

This document outlines the architecture for a custom Shopify storefront to sell **Chainmail's** white-label boxed IoT weather sensors and automate onboarding for new nodes joining the parametric insurance grid.

1. Product Catalog Architecture

SKU	Model Name	Role in Parametric Grid	Price
S-100	Sentinel Gateway Hub	Central Wi-Fi receiver dongle. Uploads signed payload streams to ingestion APIs.	\$49.00 USD
S-700	Sentinel CAT-7 Array	Solar-powered 7-in-1 array with ultrasonic wind speed and rain sensors. Zero moving parts.	\$199.00 USD
S-WH51	Sentinel Soil Probe	Wireless capacitive soil sensor to evaluate flood saturation.	\$24.00 USD
S-WFC01	Sentinel Smart Valve	Water timer valve. Auto-wets grounds on parametric dry-soil warnings (wildfire defense).	\$79.00 USD

2. Automated Onboarding Sequence

- Checkout:** Customer buys a Sentinel kit on the Shopify subdomain.
- Webhook:** Shopify posts an `order/paid` event containing customer and shipping address data to our API.
- Geocoding:** API verifies the webhook signature, calls Google Maps API for rooftop coordinates, and saves the property in Firestore as `PENDING_ACTIVATION`.
- Key Generation:** System generates a unique activation token, emailing a setup QR code to the customer.
- Activation:** When plugged in, the device sends telemetry to `/api/ingest/ecowitt`. Backend matches the token and MAC address, updating the state to `VERIFIED_NODE` and activating coverage.